MIL-D-46486F (PA)
13 February 1976
SUPERSEDING
MIL-D-46486E (MU)
22 August 1973

MILITARY SPECIFICATION

DELAY ELEMENT, M2, PARTS FOR, LOADING, ASSEMBLING AND PACKING

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing for one type of delay element designated as Delay Element M2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-650 - Explosives: Sampling Inspection and Testing
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKAGING DRAWINGS

9311702 - Delay Element, M2, Assembly
9258182 - Box, Set-up, Packing, Ammunition, Inner, for Small Explosive Components
9258183 - Box, Set-up, Packing Ammunition, Outer, for Small Explosive Components
9258184 - Box, Packing, Ammunition, for Small Explosive Components

INSPECTION EQUIPMENT DRAWINGS

8816014 - Ball, Steel
8816367 - Guide, Firing Pin
9200012 - Pin, Firing
9200602 - Bushing
9247134 - Disc, Centering
9247135 - Assembly, Firing

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and Assemblies.-The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9311702, and with all requirements specified in applicable specifications.

3.3 Moisture Content.-The moisture content of the following explosives at the loading station at the time of loading the delay elements shall not exceed the percentage specified in Table I.

TABLE I

Explosive	Percentage	Test Procedure
Primer Mixture NOL #130	0.30 maximum (max.)	4.5.2.1
Delay Composition	0.80 max.	4.5.2.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

3.4 FUNCTIONING

3.4.1 Time.-The delay element shall function high order on the first attempt under an applied energy of 15.68 inch-ounces, within the time limits specified in Table II.

TABLE II

Delay Time (Sec.)	Lower Limit (Sec.)	Upper Limit (Sec.)
0.050	0.025	0.065

3.4.2 Duds.-The delay element shall not dud. Delay elements that fail to fire on the first attempt are duds.

3.5 Waterproofness.-The delay element shall comply with the ambient requirements of 3.4.1 and shall not dud after being submerged in water to a minimum (min.) depth of two (2) inches for not less than 48 hours, at a temperature of 75 degrees (°) Fahrenheit (F) plus or minus (±) 5°F.

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.7 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility For Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing	Quantity
Body, Delay	9282910	50
Baffle	9282911	50
Cup, Primer	9282914	50
Disc, Closing	9282913	50
Primer, Stab Assembly	9282912	50
Delay, Element	9311702	385

4.3.2 Inspections to be Performed.-See MIL-A-48078, and Table III specified herein.

4.3.3 Rejection.-(See MIL-A-48078).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

2878

TABLE III - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46486F (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Delay Element, M2, Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Delay Element, M2, Assembly (Dwg. 9311702)				
	Examination for Defects	(a)		3.2	4.4.2.8
	Functioning	315(c)		3.4	4.5.4 (b)
	Waterproofness	70(d)		3.5	4.5.5 (b)
	Body, Delay (Dwg. 9282910)				
	Examination for Defects	50		3.2	4.4.2.1
	Baffle (Dwg. 9282911)				
	Examination for Defects	50		3.2	4.4.2.2
	Cup, Primer (Dwg. 9282914)				
	Examination for Defects	50		3.2	4.4.2.3
	Disc, Closing (Dwg. 9282913)				
	Examination for Defects	50		3.2	4.4.2.4
	Primer, Stab Assembly (Dwg. 9282912)				
	Examination for Defects	50		3.2	4.4.2.6

NOTES:
(a) all samples from functioning & waterproofness to be examined prior to functional testing
(b) see Special Inspection Equipment Drawings
(c) duds 315-1-2, delay times 315-5-6 (d) waterproofness 70-0-1

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of detonators shall contain:

a. Metal parts from not more than one lot interfix number from one supplier.

b. Primer mix NOL #130 or delay composition. Each explosive from not more than one batch or blend of batches which results in homogeneity of each explosive used in loading a lot.

c. M-7 relays from not more than one lot interfix number from one supplier.

4.4.2 Examination.-(See MIL-A-48078).

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46486F (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Body Delay		9282910
			NEXT HIGHER ASSEMBLY 9311702

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of outside diameter to pitch diameter		0.40%	3.2	Gage
102	Inside diameter of delay charge cavity		0.40%	3.2	Gage
103	True position of primer cavity to outside diameter		0.40%	3.2	Gage
104	True position of relay cavity to outside diameter		0.40%	3.2	Gage
105	Pitch diameter, minimum (min)		0.40%	3.2	Gage
106	Major diameter, min		0.40%	3.2	Gage
Minor					
201	Length, min		0.65%	3.2	Gage
202	Outside diameter, min		0.65%	3.2	Gage
203	Inside diameter of relay cavity, max		0.65%	3.2	Gage
204	Inside diameter of primer cavity, max		0.65%	3.2	Gage
205	Depth of relay cavity, max		0.65%	3.2	Gage
206	Depth of primer cavity, max		0.65%	3.2	Gage
207	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46486F (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9282911
4.4.2.2	Baffle				NEXT HIGHER ASSEMBLY 9311702
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness		0.40%	3.2	Gage
102	Diameter of flash holes		0.40%	3.2	Gage
Minor					
201	Outside diameter		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

Source: https://assist.dla.mil - Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46486F (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9282914
4.4.2.3	Cup, Primer				NEXT HIGHER ASSEMBLY 9282912
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness through coined surface		0.40%	3.2	Gage
102	Diameter of coined surface, min		0.40%	3.2	Gage
Minor					
201	Length		0.65%	3.2	Gage
202	Outside diameter		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46486F (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Disc, Closing		9282913
			NEXT HIGHER ASSEMBLY 9282912

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter, min		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

10

Source: https://assist.dla.mil/ Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46486F (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9282912
4.4.2.5	Primer, Stab Assembly (prior to loading of primer mixture)				NEXT HIGHER ASSEMBLY 9311702
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture content of primer mixture	(a)		3.3	4.5.2.1
102	Composition of primer mixture	(a)		3.1	4.5.3
Minor	None defined				
NOTES: (a)	The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one (1) sample of each eight (8) hours of production. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, that quantity of material represented by the sample shall be rejected. If primer, stab assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample, together with all primer, stab assemblies loaded with the nonconforming material, shall be rejected. The tests shall be performed as specified in 4.5.2.1 or 4.5.3 as applicable.				

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46486F (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Primer, Stab Assembly				9282912
					NEXT HIGHER ASSEMBLY
					9311702
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Cup split, cut or cracked		100%	3.2	Visual
Major					
101	Outside diameter, max		0.40%	3.2	Gage
102	Length, max		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46486F (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Delay Element, M2, Assembly (Prior to loading of delay composition)				9311702
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Moisture content of delay composition	(a)		3.3	4.5.2.2
NOTES:(a)	The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one (1) sample of each eight (8) hours of production. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement and loading has not begun, that quantity of material represented by the sample shall be rejected. If delay elements have been loaded with material containing excessive moisture, the remaining material that is represented by the sample, together with all delay elements loaded with the nonconforming material, shall be rejected. The tests shall be performed as specified in 4.5.2.2.				

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46486F (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Delay Element, M2, Assembly				9311702
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, max		0.40%	3.2	Gage
102	Diameter, max		0.40%	3.2	Gage
103	Depth to primer firing surface, min		0.40%	3.2	Gage
104	Primer inverted		0.40%	3.2	Visual
105	Lacquer seal missing, either end		0.40%	3.2	Visual
106	Crimp not full 360°		0.40%	3.2	Visual
107	Functioning	(a)		3.4	4.5.4
108	Waterproofness	35(b)		3.5	4.5.5
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: (a) See 4.5.4 - duds - AQL 0.15% - delay times - AQL 0.65%
(b) Waterproofness 35-0-1

Source: http://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46486F (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Box, Set-up, Inner (after loading and prior to closing slide cover)				9258182
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Filler and cushion assembly missing		100%	3.2	Visual
2	Filler and cushion assembly improperly assembled		100%	3.2	Visual
3	Stiffener missing		100%	3.2	Visual
4	Slide cover assembled reversed		100%	3.2	Visual
Major					
101	Release paper missing		0.40%	3.2	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

Source: https://assist.dla.mil - Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46486F (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Box, Set-up, Outer (with inner set-up boxes assembled and prior to assembling top)	1 OF 1	9258183
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up boxes		100%	3.2	Visual
2	Stiffener not facing one direction		100%	3.2	Visual
Major	None defined				
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-D-46486F (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Box, Packing Ammunition (prior to assembling cover and strapping)	1 OF 1	9258184
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bags containing set-up boxes damaged		0.40%	3.2	Visual
102	Cushioning material incomplete or missing from around outer set-up boxes		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4B10

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-D-46486F (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Box, Packing, Ammunition				9258184
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Strapping missing, broken, loose or improperly fastened		0.40%	3.2	Visual
102	Box damaged		0.40%	3.2	Visual
103	Marking missing, broken		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.-Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the First Article or Quality Conformance Inspection tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test Methods and Procedures

4.5.1 Materials, Components and Processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspections.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table III.

4.5.2 Moisture Content

4.5.2.1 Primer Mixture NOL #130.-The moisture content of the primer mixture shall be determined in accordance with Method 101.6 of MIL-STD-650.

4.5.2.2 Delay Composition.-The moisture content of the delay composition shall be determined in accordance with Method 101.5 of MIL-STD-650.

4.5.3 Composition of primer mixture NOL #130 shall be determined as follows:

4.5.3.1 Primer Mixture NOL #130

4.5.3.1.1 Barium Nitrate.-An accurately weighed portion of approximately 0.5 gram (g) of primer mixture shall be transferred to a tared, medium porosity, fritted glass crucible of 20 milliliters (ml) capacity. Three (3) ml of lead azide saturated distilled water shall be added at a temperature of 50° Centigrade (c), ± 2°C, agitated for one (1) minute, and if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five (5) times, making a total of six (6) 3-ml extractions. The sample in the filtering crucible

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

shall be rinsed three (3) times with alcohol, dried in an oven maintained at 60°C ± 5°C for thirty (30) minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barium Nitrate, Percent} = \frac{M\ (A-B)\ 100}{1-0.01M}$$

where:

A = Weight of crucible and sample
B = Weight of crucible and sample after extractions with lead azide saturated water
M = Moisture content of sample obtained from 4.5.2.1 in percent

4.5.3.1.2 Basic Lead Styphnate (when applicable).-The residue obtained in the determination of barium nitrate shall be extracted with six (6) ml portions of saturated ammonium acetate solution at about 25°C (hot saturated ammonium acetate extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. Reserve the crucible and residue for the tetracene determination. The filtrate and washings shall be quantitatively transferred to a 250-ml calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2-ml aliquot of this solution shall be transferred to a 50-ml calibrated volumetric flask and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrophotometric cell having a width of approximately 1 centimeter (cm), and its absorbance determined at a wave length of 410 millimicrons. As reference, a similar cell containing saturated ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbance obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated on a dry basis as follows:

$$\text{Basic Lead Styphnate, Percent} = \frac{F\ (A-B)}{W\ (1-0.01M)\ D}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

where:

A = Absorbance of basic lead styphnate solution
B = Absorbance of ammonium acetate solution
D = Width of corex cell, cm
M = Moisture content of sample obtained from 4.5.2.1 in percent
W = Weight of sample
F = Spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantitively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.3.1.3 Determination of Normal Lead Styphnate (when applicable).-Extract the residue obtained in the barium nitrate determination with sufficient 5-ml portions of ammonium acetate solution (saturated) at about 25°C until the washings are colorless. Agitate each portion for about 1 minute, then allow each portion to remain in contact with the residue for 1 additional minute. Finally, wash the residue with water, saturated with antimony sulfide, until the washings come through clear. Reserve the crucible and residue for the tetracene determination. Transfer the filtrate and washings quantitatively to a 250 ml calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark with distilled water. Using a calibrated pipette, transfer a 2 ml aliquot of this solution to a second 100 ml calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell having a width of 1 cm and determine its absorbance at a wave lengh of 410 millimicrons. Use as reference, a similar cell containing saturated ammonium acetate solution of the same concentration as the sample in the final dilution. Since the cell which holds the normal lead styphnate is not identical with the reference cell, correct the absorbance obtained as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, fill both cells with ammonium acetate solution and measure at a wave length of 410 millimicrons, the absorbance of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the sample on a dry basis as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

$$\text{Normal Lead Styphnate, Percent} = \frac{F\,(A-B)}{WD}$$

where:

A = Absorbance of normal lead styphnate solution
B = Absorbance of ammonium acetate solution
D = Width of corex cell, cm
W = Weight of the dried sample
F = Spectrophotometric factor. To determine the spectrophometric factor (F) weigh a 0.5 gm sample of pure lead styphnate. Transfer the lead styphnate quantititatively with 50 to 100 ml of distilled water to a 500 ml volumetric flask, add 30 ml of saturated ammonium acetate solution to the solution in the flask and dilute to mark with distilled water. Using aliquots of the solution prepare five solutions of lead styphnate ranging in concentration from .5 to 2.5 mg per 100 ml. Determine the absorbance of each solution at 410 millimicrons. Divide concentrations of lead styphnate for each of the solutions by its corresponding absorbance value. The average of the five quotients thus obtained shall be used as the spectrophometric factor (F) for the lead styphnate.

4.5.3.1.4 Tetracene.-The wet residue obtained in the determination of lead styphnate shall be washed three (3) times with alcohol, dried in an oven maintained at 60°C ± 5°C for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125-ml beaker with a stream of water; 25 ml of water shall be added and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three (3) times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60°C ± 5°C for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

$$\text{Tetracene, Percent} = \frac{(C-D)\ 100}{W\ (1-0.01M)}$$

where:

C = Weight of crucible and residue after extraction with saturated ammonium acetate solution
D = Weight of crucible and residue after treating with hot water
W = Weight of sample
M = Moisture content of sample obtained from 4.5.2.1 in percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

4.5.3.1.5 Antimony Sulfide.-The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

$$\text{Antimony Sulfide, Percent} - \frac{(D-E)\ 100}{W\ (1-0.01M)}$$

where:

D = Weight of crucible and sample after treating with hot water
E = Weight of crucible
W = Weight of sample
M = Moisture content of sample obtained in 4.5.2.1 in percent

4.5.3.1.6 Lead Azide.-The lead azide content in the sample, on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, lead styphnate, tetracene and antimony sulfide.

4.5.4 Functioning

4.5.4.1 Delay Time and Duds.-The delay element shall be assembled to the test fixture and a steel ball (dwg. 8816014) weighing 1.96 ounces, max. shall be dropped from a height of 8 inches max. onto the cap of the firing pin (dwg. 9200012). The height shall be measured from the bottom of the steel ball to the top of the firing pin. The tests shall be performed using equipment in accordance with 4.4.4 and dwgs. 8816014, 8816367, 9200012, 9200602, 9247134 and 9247135. The elapsed time between the ball striking the firing pin and the detonation of the relay shall be measured to determine compliance with the requirement.

4.5.5 Waterproofness.-The delay element shall be conditioned as specified in 3.5, then removed and subjected to the functioning test as specified in 4.5.4.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.-The detonators shall be packaged and marked in accordance with dwgs. 9258182 and 9258183.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

5.1.2 Level C.-The detonator parts shall be packaged in accordance with MIL-STD-1169 requirements.

5.2 Packing

5.2.1 Level C.-The detonators shall be packed and marked as specified on dwg. 9258184.

5.2.2 Level C.-The detonator parts shall be packed in accordance with MIL-STD-1169 requirements.

5.3 Marking.-Detonator containers and bags shall be marked in accordance with instructions contained in dwgs. 9258182, 9258183, and 9258184. Detonator parts containers shall be marked in accordance with MIL-STD-1169 requirements.

6. NOTES

6.1 Ordering Data.-See MIL-A-48078 (PA).

6.2 Submission of Inspection Equipment Designs for Approval (See MIL-A-48078).-Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T, Dover, New Jersey 07801.

6.3 Check Test For Possible Deterioration.-The following test is designed to prevent delay elements that have deteriorated from being assembled into major ammunition items. The test shall be performed on delay elements from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the delay elements are assembled does not require that the check test be performed, the Contracting Officer should insert in the contract a provision for the performance of the check test. The Contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration. Delay elements shall be selected from each lot for this test.

6.3.1 Output.-A sample of 200 delay elements shall be selected at random from each lot. If two (2) or more delay elements dud or four (4) or more delay elements fail to comply with the requirement for the delay time the lot shall be rejected.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1375-A-166

Source: https://assist.dla.mil -- Downloaded: 2020-10-27 06:04Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDER
Picatinny Arsenal
ATTN: SARPA-QA-A
Dover, New Jersey 07801

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:04Z
Check the source to verify that this is the current version before use.